D02 · S&OP; / IBP Governance

# 05 · S&OP; maturity model & KPI framework

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## 1 WHAT IT IS

An S&OP maturity model and KPI framework measure the health of the planning process itself — not only its outcomes — across defined maturity stages, using process metrics (forecast accuracy, bias, plan adherence, FVA, meeting effectiveness) alongside outcome metrics (service, inventory, cost), so the organization can see where it is, what the next stage requires, and whether each cycle is improving.

## 2 WHY IT MATTERS

Grimson & Pyke 2007 **(Grimson & Pyke, 2007 — peer-reviewed)** built an empirically grounded S&OP maturity framework across five dimensions — meetings and collaboration, organization, measurement, information technology, and S&OP plan integration — showing that maturity is multidimensional and advances unevenly. An organization that measures only outcomes (service, inventory) cannot tell whether a good month came from a good process or from luck, nor diagnose why a bad month happened. Without process KPIs, S&OP cannot improve deliberately: it has no instrument panel, only a rear-view mirror. The result is the common plateau — **a process stuck at low maturity for years**, busy every month yet never getting better, because nothing measures the process itself.

## 3 HOW IT IS DONE

1. **Baseline current maturity across dimensions.** Assess the process on the recognized dimensions — meetings/collaboration, organization, measurement, technology, plan integration — to locate where it actually sits, not where it claims to. Maturity advances unevenly; the baseline shows which dimension is the binding constraint.
2. **Separate process KPIs from outcome KPIs.** Outcome metrics (fill rate, inventory, cost) tell you what happened; process metrics (forecast accuracy, bias, plan adherence, FVA, meeting effectiveness) tell you why. A scorecard with only outcomes cannot diagnose, only report.
3. **Instrument forecast accuracy and bias separately.** Accuracy and bias are different failures: accuracy is noise, bias is a systematic lean. A process can be accurate-but-biased or unbiased-but-noisy, and the corrective action differs — track both, not a single blended error number.
4. **Measure plan adherence and FVA.** Plan adherence asks whether the committed plan held; FVA asks whether each step of the process added or destroyed accuracy. Together they reveal whether the process is a control loop or a ritual that touches the number without improving it.
5. **Define the target stage and its required capabilities.** Pick the next maturity stage and list the concrete capabilities it requires — a supply-balancing step, a financial bridge, scenario triggers. Maturity is a roadmap of capabilities to build, not a score to admire.
6. **Review maturity and KPIs on a fixed cadence.** Re-assess maturity periodically and track the process KPIs every cycle. A maturity model assessed once and shelved is a slide; assessed continuously, it is the instrument that drives deliberate improvement.

**WORKED EXAMPLE — ILLUSTRATIVE**

A company measured S&OP only by service and inventory — **baseline: no process KPIs, maturity self-assessed as "advanced" but plateaued, recurring misses no one could explain**. It baselined maturity across the five dimensions (revealing a missing supply-balancing step and no FVA), added forecast accuracy, bias, plan adherence, and FVA to the scorecard, and set a quarterly maturity review. Results: **forecast bias surfaced at +9% and corrected to +2%, plan adherence 71% -> 88%, and a clear capability roadmap that moved the weakest dimension up one stage in two quarters**.

**IN PRACTICE**

### A balanced S&OP scorecard

Build one scorecard that pairs outcome metrics with process metrics, so leadership sees service and inventory next to forecast accuracy, bias, plan adherence, and FVA. The pairing forces the diagnostic question — a good outcome with a bad process is luck, and a bad outcome with a good process is signal.

### Maturity as a capability roadmap

Translate the target maturity stage into a concrete list of capabilities to build — the supply-balancing step, the financial bridge, scenario triggers, master-data governance. Maturity scoring is only useful if it produces a roadmap; a score with no next action is vanity.

### Track bias, not just accuracy

Monitor forecast bias explicitly and persistently. A persistent positive or negative bias is the most fixable forecast error and the most expensive if ignored — it systematically over- or under-positions inventory every single cycle until someone measures and corrects it.

### Periodic maturity re-assessment

Re-assess maturity on a regular cadence and compare against the prior baseline. This converts the maturity model from a one-time consulting deliverable into a living management instrument, and keeps the improvement roadmap honest about what actually advanced versus what was merely planned.

#### WHAT YOU WOULD USE

**S&OP MEASUREMENT / KPI & MATURITY**

**SAP IBP / o9 — embedded analytics** — Native forecast-accuracy, bias, FVA, and plan-adherence analytics on the planning data, with configurable S&OP dashboards. **(SAP, o9 — vendor)** The choice when process KPIs should live on the same platform that produces the plan, with no separate data pull.

**Kinaxis / Blue Yonder analytics** — Strong embedded planning analytics and KPI dashboards covering accuracy, adherence, and exception trends. **(Kinaxis, Blue Yonder — vendor)** A solid fit where the measurement layer rides on the existing planning engine.

**BI / analytics on a planning data mart** — Power BI, Tableau, or similar built on a planning data mart, giving full control over FVA and bias definitions. **(BI vendors — vendor)** The vertical option when the KPI framework must be tailored beyond what the APS reports natively.

**For operations that don't need a platform** — A maturity self-assessment against a published framework plus a spreadsheet scorecard tracking accuracy, bias, plan adherence, and FVA each cycle. The discipline of measuring the process — and acting on a capability roadmap — matters far more than the dashboard that displays it.

## 4 SO WHAT

> **Operations that measure the process improve it deliberately; operations that measure only the outcome can tell you the month was bad but never why, and so repeat it.** The S&OP plateau is not a mystery — it is what happens when a process has no metric for its own health. Outcome KPIs grade the result; process KPIs and a maturity model are the only instruments that turn a recurring meeting into a system that compounds.

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